



MUSIC IS NUMBERS

By Cynthia Jordan

cynthiamusic.com

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*To my teacher
Doctor Nino Albanese,
my friend and mentor*

Dear Music Lover,

Thank you for joining me on your journey with music. I have developed this easy program with the intention to teach the simple mechanics of playing and creating music on the keyboard. I will show you how you can create your own compositions, play your favorite songs and even play in a band.



We live in a 3 dimensional world measured by numbers. Music is also 3 dimensional and defined with numbers. The three dimensions are note to note and the space in between. Anyone who can count to 7, can play music.

Music is the universal language that speaks to all living things. Just like breath, the beating of our hearts and waves breaking on the sand, it is the rhythm that evokes the emotion of the music. Timing is dictated in the space between the notes. All of the chord structures and scales are formed with simple patterns. These harmonic sounds can evoke happy tones (major) as well as dramatic tones (minor). It is the harmonics and timing that tells a story with music.

Music is energy. There is evidence that plants grow healthier when exposed to uplifting music. Some farmers have been known to put speakers in their fields to play symphonies for their crops. Many times when I play the piano, small animals will come on my porch to be entertained. A lullaby with the “ahh” sound, will instantly calm a baby and put it to sleep. Even the earth sings as she soars through the solar system spinning on her axis.

Artists paint on canvass. Composers create on silence. Much like yellow and blue make green, I will teach you how to combine notes to make beautiful harmonic sounds. Soon be able to create a style of playing that s unique only to you. To watch our teaching videos, visit us at cynthiamusic.com. Congratulations. Now let's have some fun.

*Many thanks!
Cynthia*

"Music directly imitates the passions or states of the soul... when one listens to music that imitates a certain passion, he becomes imbued with the same passion; and if over a long time he habitually listens to music that rouses ignoble passions, his whole character will be shaped to an ignoble form."
~Aristotle

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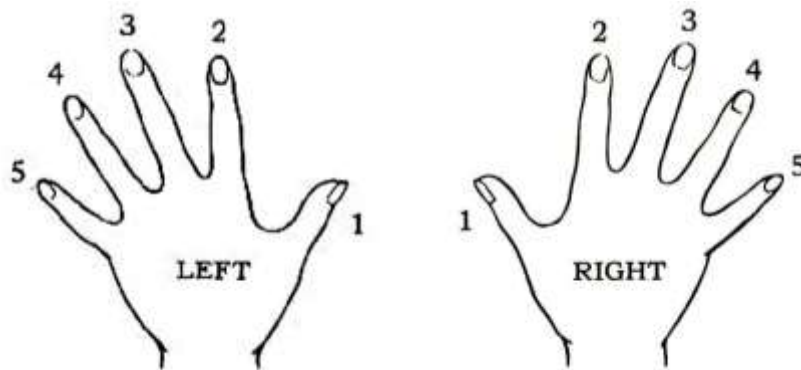
Learning the Keyboard

Fingering

Most music is written with fingering. The small numbers written under the notes indicate which finger to use while playing. This makes it easier to play so that your fingers don't get all tangled up. You will learn more about this when you learn how to read music.

1= thumb 2= pointer 3= middle 4= ring 5= pinky

Here is a great exercise you can do while watching TV.

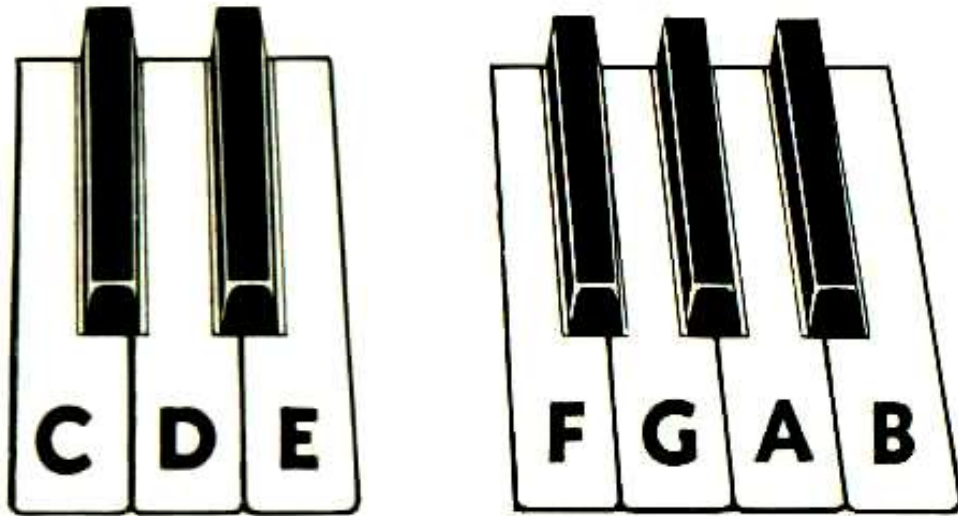


1. Lay your hand flat, then slightly curve your hand up gripping with your fingertips and keeping the pad of your hand flat on the table (or your legs), Put down your thumb, then your middle finger then your pinky keeping your 2 and 4 fingers up.
2. This time put down your thumb, then your pointer, then your pinky. Repeat, keeping fingers 3 and 4 up. Alternate placing each finger one at a time or all together.
3. Now add the pinky and put down 1 – 2 – 3 – 5

Many people have a tendency to round their hands like claws. Don't do this! It is a hard habit to break. Relax your hands and shoulders. Once you have mastered these two hand positions as second nature you will be amazed at what you will be able to play.

Just 7 letters!

In this program, we will be working with the Key of C because it is the key with no flats or sharps. This means we will only be working with the white piano keys.



The piano keyboard is divided into 2 groups as shown above. They are the group of 2 black keys and the group of 3 black keys.

The CDE group is easy to remember because it is just 3 letters. C is in the front, D is in the middle and E is in the back.

In the FGAB group, there are 2 white notes in the middle. When I was a little girl I imagined that the group of 3 black keys was a house. The “F” is the front door and the “B” is the back door. Inside is a baby who says “GA”. It works for me!

Exercise:

Locate all the C’s on the piano.

The F’s... the A’s... the D’s...

the B’s... the G’s and now finally, the E’s.

Finding Middle C

K.Kawai



Middle C is the C located in the middle of the piano. It is the C directly under the name of the make of the piano. My piano is a K.Kawai and the picture above illustrates how I view Middle C on my piano keyboard.



The music staff consists of 5 lines and 4 spaces. Above is how Middle C is written in music. All the notes you see are Middle C. Middle C is like the zero in numbers. It is the center point used to separate the keyboard to make music easier to read.

The staff above is the treble clef and the staff below is the bass clef. As you can see, if there was a line between the two staves it would be Middle C. When the stems are up you play with your right hand and when they are down you play with your left hand.

The vertical lines are bar lines that separate the measures. In the music above each measure has 4 beats. In the first measure, each note is a quarter note and is held for one count. In the second measure, each note is a half note and held for 2 counts. The last measure has a whole note which is held for 4 counts.

Playing the C Scale

	C	D	E	F	G	A	B	C
Fingers to use	1	2	3	1	2	3	4	5

To play the C Scale going up

Begin by putting the thumb of your right hand on Middle C.

Release as you play D with your 2 finger.

Release while you play E with your 3 finger.

With your 3 finger remaining on the note, rotate your hand to the left and cross your thumb UNDER the 3 finger.

Play F with your thumb.

Release as you play G with your 2 finger.

Release as you play A with your 3 finger.

Release as you play B with your 4 finger.

Finish while playing the C above Middle C with your 5 finger.

To play the C Scale going down

Begin by playing the C above Middle C with the 5 finger of your right hand.

Release as you play B with your 4 finger.

Release as you play A with your 3 finger.

Release as you play G with your 2 finger.

Release as you play F with your thumb.

With your thumb remaining on the note, rotate your hand to the left and cross your 3 finger OVER your thumb.

Play the E with your 3 finger.

Release as you play D with your 2 finger.

Finish right where you began with your thumb on Middle C.

Place your pinky of your left hand on the C below Middle C and play the C Scale with your left hand. The fingering is 5 – 4 – 3 – 2 – 1 – 3 – 2 – 1 going up and then beginning with Middle C use 1 – 2 – 3 – 1 – 2 – 3 – 4 – 5 going down.

Musical Tones Are Like Colors

We are going to work with the C scale.

C	D	E	F	G	A	B
1	2	3	4	5	6	7
Major	minor	minor	Major	Major	minor	diminished

I like to think of the 1, 4 and 5 or in the C scale C, F, and G as primary colors (red, blue and yellow). These are your major chords.

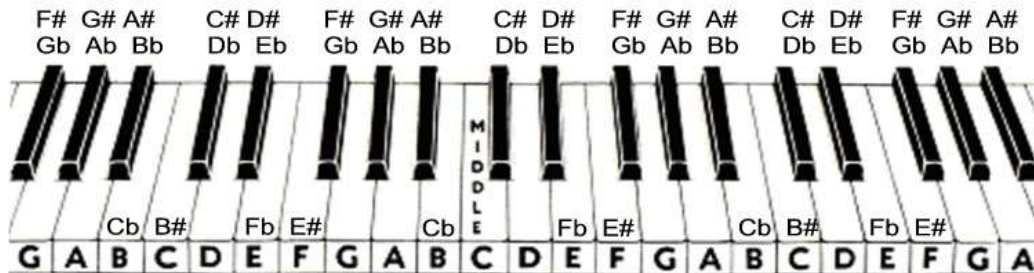
The 2, 3 and 6 or the D, E and the A are like the secondary colors (orange, green and purple). These are the minor chords written Dm, Em, and Am.

When playing the C chord play the C with your thumb, the E with your middle finger and the G with your pinky. Use the same fingers for all the chords. Use this chart to get familiar with the different chords in the C scale.

C	C (1 thumb)	E (3 middle)	G (5 pinky)
Dm	D	F	A
Em	E	G	B
F	F	A	C
G	G	B	D
Am	A	C	E

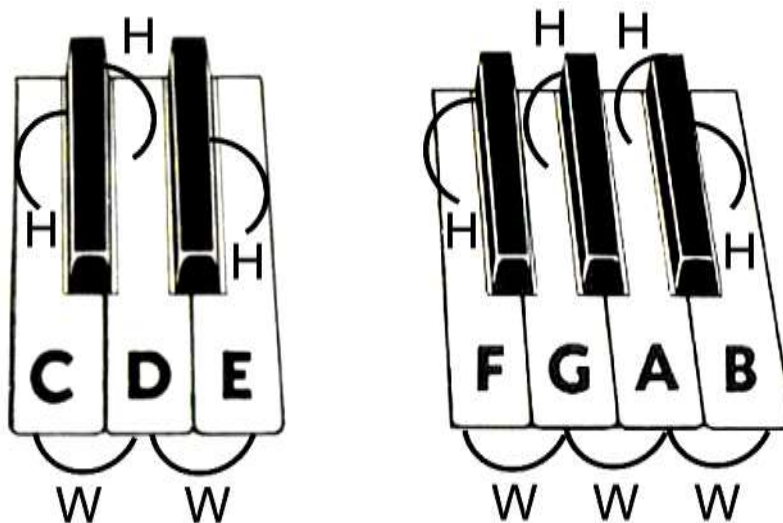
In the C scale, the B is diminished chord and has a different kind of sound usually used in Jazz. It will be learned at a later date.

Half and Whole Steps



To complete the understanding of the black and white notes on the keyboard it is important to know the difference between half and whole steps. Half steps are 1 tone apart and whole steps are 2 tones apart.

Look at the diagram. You can see that Gb (G flat) and F# (F sharp) are the same note. Flats (bs) are always a half step lower than the white tone indicated. Sharps (#s) are one-half step above the note indicated. They are not necessarily black. Notice for instance, that B# is also a C.



The W's show whole tones and the H's show the half steps. C to C# is a half step. C to D is a whole tone etc.

Playing Octaves

An octave is the same note 8 tones apart. Playing an octave is playing two tones simultaneously with the same name. To play a C octave, play Middle C with your left thumb and the C below it with your left pinky simultaneously. Now play Middle C with the thumb of your right hand and the C above it with your right pinky simultaneously.

Try playing the C Scale using octaves and then play various octaves all over the keyboard naming the note out loud as you play each one.

Try this:

1. Play the C below Middle C with your left thumb and the C below that with your left pinky simultaneously. This is a nice full sounding C octave.
2. Hold it down as you play the C scale with your right hand. Press the sustain pedal (the furthest pedal on the right) to create an ambiance. Play the notes slow and dramatically.

The C scale going down is the same sequence of notes as the Christmas Carol “Joy to the World”. Try it with the rhythm that follows the song. As you play the C scale going down sing the words “Joy to the world, the Lord has come” in your head and play along.

Now, while holding down an A octave, play the white notes in any order with the different rhythms of our world. For instance, as you play, think of the rhythm of a heartbeat, walking, running, rain, breathing, birds etc.. Long pauses between the notes can be beautiful and dramatic. Create a mood as you play.

PIANO PEDALS

The pedal on the far right is the SUSTAIN PEDAL. When you press down on it makes the notes ring. It is a good idea to clear it every chord change or it will sound muddy.

The far left pedal is the SOFT PEDAL. When pressed down the hammer hits only 2 strings instead of 3.

The middle is the SOSTENUTO PEDAL. It also sustains but is rarely used.



Compose Your Own Music

Now it's time to be *really* creative. Play that same C octave with your left hand and push down the sustain pedal. With your right hand play the notes of the C chord at random which are the notes C, E and G. Play them in any order. Hold them down as long as you wish. Remember to make dramatic pauses.

Try it with all of the chords using the notes of the chord in any order, any sequence, and in any rhythm pattern. Try different combinations and remember them. These are *your* creations!

Left hand (Bass)	Notes to use	Chord
C octave	C – E – G	C
D octave	D – F – A	Dm
E octave	E – G – B	Em
F octave	F – A – C	F
G octave	G – B – D	G
A octave	A – C - E	Am

Here is a creative exercise I like to do. With your left hand play the lower A octave. Now play the G octave right next to it. Hold your pedal down but clear it with an up-down motion when you change your octave. Otherwise, it will get too muddy sounding.

As you alternate back and forth from A to G, play all white notes in your right hand in any order. Find different patterns you like and remember the notes that work best. After you get comfortable with A and G you can add the F octave. Find combinations *you* like. Let the music ring! Remember the drama is in the space between the notes.

The A minor Scale

I chose to teach you with the Key of C because it only uses the white keys on the piano. It is also the Key of A minor. This is because A minor is the Relative Minor to C. Songs written in minor keys can sometimes sound a bit sad but they are so beautiful! For instance, the song *Greensleeves* is written in a minor key.

The Am Scale is:

A B C D E F G A

The Relative Minor of every major chord is the chord that most compliments the chord and sometimes can even be used instead of the major chord. This works well especially with the 4 chord and its relative minor which is the 2 chord of the scale. For instance,, when playing in the key of C you can substitute Dm for F to create a slightly different sound.

All of the major chords have their relative minors present in every scale.

We learned that in the C scale the 6 harmonic chords are:

C Dm Em F G Am

Am = relative minor to C

Dm = relative minor to F

Em = relative minor to G

Isn't it amazing to see how beautifully harmonics work together in the Music System? It's all about numbers and relativity. Did you know that the word "universe" means one-song?

Noodling

Doodling is when we make random pictures on scraps of paper. They don't have to make sense and they usually consist of funny shapes or images that we don't necessarily need to keep or frame. However, sometimes doodling can evoke an idea that has the potential to grow into something grand.

“Noodling” is to music as “doodling” is to pencil and paper. Some of my best compositions were conceived by “noodling” on the keyboard.

I like to noodle using the exercise I described on page 14. It is like zen to me...

Simply play A, F, and G octaves with your left hand in any order. Repeat them, alternate them and find patterns you like.

With your right hand play any white note. Again, repeat them, alternate them, and find patterns you like.

When you play something you really like, name it, memorize it, even record it. When you do you can call yourself, “a composer.”

Making music is fun! I like the app VOICE RECORD. You can record your compositions and even email them to friends and family.

Please join our music community at cynthiamusic.com for more lessons, sheet music, and tutorials.

The Next Step

Before you move on may I suggest you get familiar with the six chords on page 14.

Feel each individual chord in your fingers and SAY EACH NOTE OUT LOUD as you play it.

.

Break the chord up, that is to play each note of the chord one finger at a time.

Play the chords in different locations on the keyboard and SAY EACH NOTE OUT LOUD.

Play each chord from the octave position. Let's begin with the C chord.

With the right hand, play Middle C with your thumb, the E with your second finger, the G with your third and the top C with your fifth finger.

With the left hand, your fifth finger will be on the lower C, your third finger on the E, your second finger on the G and your thumb on the high C.

Use this fingering when playing all six chords in the C Scale.

I want to remind you to keep your hands relaxed and slightly rounded as if they were laying on a soccer ball. Bend the tips slightly to act like little hammers. Play very, very slowly. Do this at least ten minutes a day. Believe me, you will thank me for this. I have spent hours and hours playing s-l-o-w-l-y. Although I didn't understand why at the time, now I can see how it has really paid off.

As you play the chords notice the notes the different chords share. For instance, the C and Am chord share the C and E. The F, and Dm chords share the F, and A, and the G and Em chords share the G and B. The F and C chord share the C, the G and C chord share the G and so on. This knowledge will serve you well.

Be creative when you practice your chords. Play different patterns. I suggest you give it a good week before you move on. Most importantly...HAVE FUN!

Playing by Ear on the Keyboard

Now that you have learned the 6 basic chords of the C scale you are ready to play thousands of songs in the key of C.

It is important to know that you are not limited to using just those 6 harmonic chords when you write music. For instance,, many times in early jazz the 2 and 3 and the 6 were used as major chords that produced a unique sound.

Transitioning from a 4 to a 4 minor is nice like in the song *Desperado* by the Eagles.

To play your favorite songs by ear in the key of C, start by playing a C chord. Find your first notes by singing them and then imitating them on the piano. Usually, you can find the harmony with the C, Dm, Em, F, G, and Am chords, however, there are exceptions.

You can play an endless amount of songs using just the 1, 4 and 5 chords. Do you know the song *La Bamba*? To play it, simply repeat the chords C, F, and G in that order over and over again. Try it. It is also the harmony to the song *Twist and Shout*.

Are you an oldies fan? Play the 1, 6m, 4, 5 sequence. In the key of C that is C, Am, F, G. This chord progression was used often in the 50s and 60s music. *Silhouettes*, *Earth Angel*, *Angel Baby*, *Heart and Soul* are all songs that use this progression.

Now try the 1, 2m, 4, 5 chord pattern. The song *Love Is All Around* from the movie, *Love Actually* and *Three Weddings and a Funeral* is played with the music progression. In the key of C that is C, Dm, F, and G.

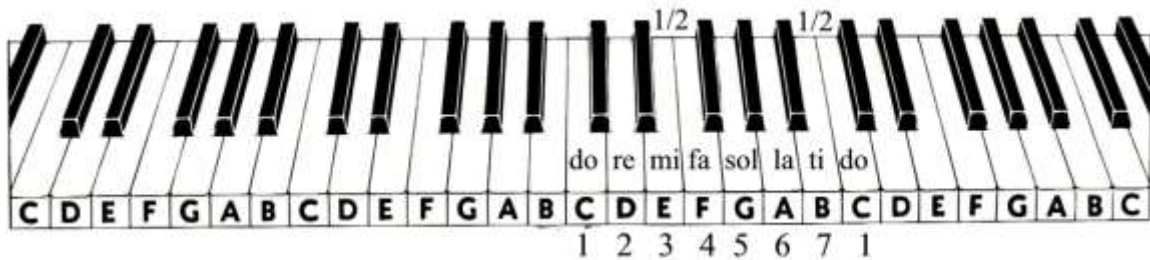
UP UNTIL NOW WE HAVE FOCUSSED ON THE KEYBOARD BECAUSE IT IS EASIEST INSTRUMENT TO LEARN PATTERNS AND MUSIC THEORY.

ALTHOUGH I WILL CONTINUE USING THE KEYBOARD TO ILLUSTRATE THE VARIOUS HARMONIC FORMULAS, THE SAME RULES APPLY TO ALL INSTRUMENTS.

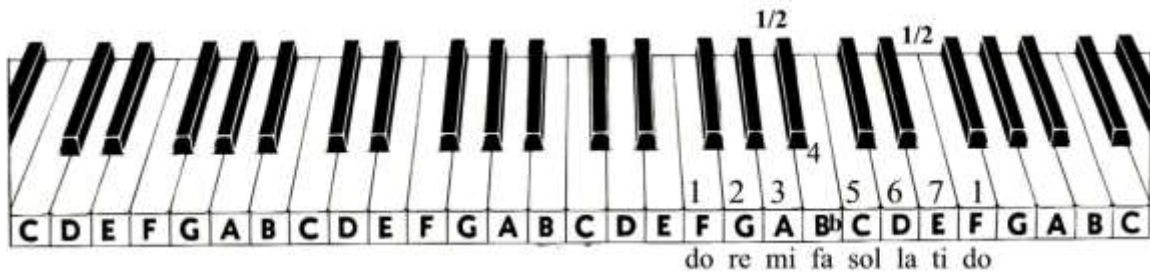
Music Formulas

To Find Any Scale

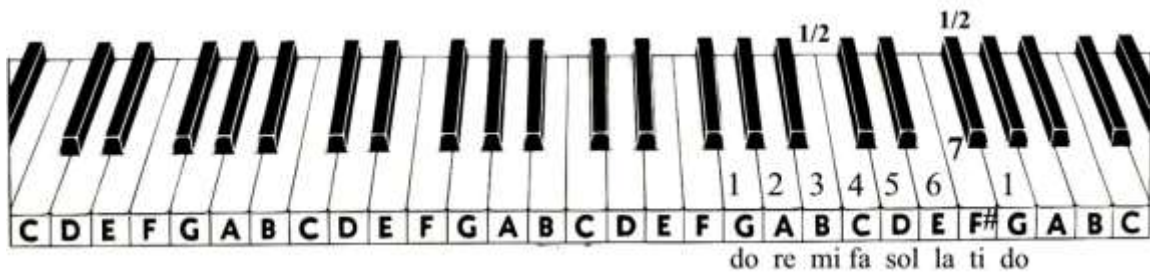
C Scale



F Scale



G Scale



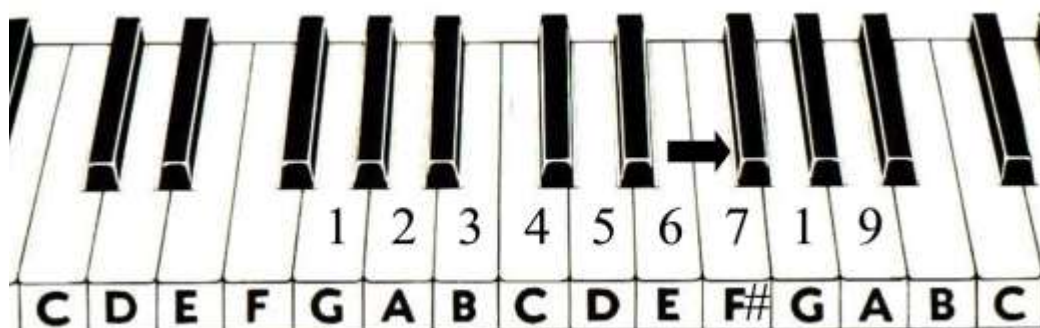
To find ANY scale

1 to 2 = whole step 2 to 3 = whole step 3 to 4 = half step

4 to 5 = whole step 5 to 6 = whole step

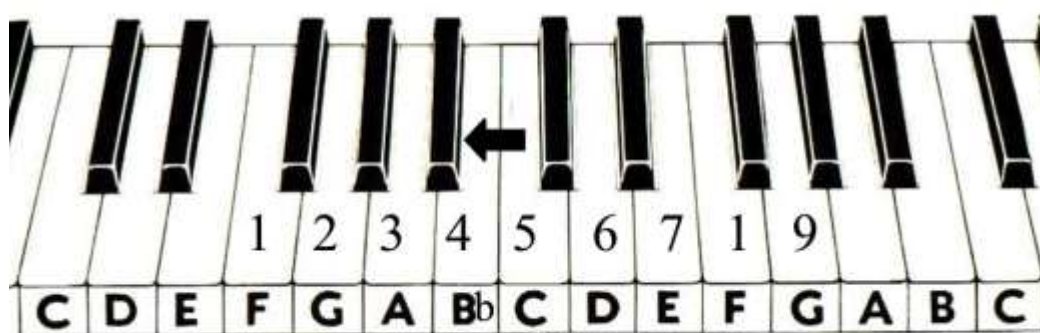
6 to 7 = whole step 7 to 8 (1) = half step

Key of G



G Am Bm C D Em F# G

Key of F



F Gm Am Bb C Dm F

The harmonic chords of every scale are determined by the notes in that scale.

The 2, 3 and the 6 chord of every scale are *always* minor.

The 1, 4 and the 5 chord of every scale are *always* major.

Scale Chart

DO	RE	MI	FA	SOL	LA	TI
1	2	3	4	5	6	7
keys C	D	E	F	G	A	B
G	A	B	C	D	E	F#
D	E	F#	G	A	B	C#
A	B	C#	D	E	F#	G#
E	F#	G#	A	B	C#	D#
B	C#	D#	E	F#	G#	A#
F	G	A	Bb	C	D	E
Bb	C	D	Eb	F	G	A
Eb	F	G	Ab	Bb	C	D
Ab	Bb	C	Db	Eb	F	G
Db	Eb	F	Gb	Ab	Bb	C
Gb	Ab	Bb	Cb	Db	Eb	F

Understanding the Scale Chart

Now let's look at the pattern. The 5 will determine the next scale and the 7th note of *that* scale will determine what note will be the new sharp.

G is the 5 and F is the 7th note. The scale with one # is G with F#.

D is the 5 of the G scale and C is the 7th note. We keep the F# and add the C#.

A is the 5 of the D Scale. The 7th note of the A scale is G. Keep the F# and C# and then add the G#.

You will find this pattern to be consistent with all of the sharp keys of which there are 7, including the C scale which has no sharps or flats. This pattern also determines the order in which you see them written in the key signature.

Now let's look at the series of the flat keys beginning with the F scale on row 11.

The pattern for the flats consists of all 4s. F is the 4th note of the C Scale. B is the 4th note of the F scale. F is the scale of one flat which is Bb.

The 4th note of the F scale is Bb. The 4th note of the Bb scale is E. Therefore, Bb is the scale with 2 flats, Bb and Eb.

The fourth note of the Eb scale is Ab. Keep the Bb and the Eb. The scale of 3 flats is Ab with Ab, Eb, and Bb.

This pattern is also consistent for all of the flat keys and the order of the flats in the key signature of a song is determined by the order in which they were added.

Key Signature – Displays which notes will be made flat or sharp.

Key Signatures - Treble Clef

 C major A minor	 G major E minor	 D major B minor	 A major F# minor	 E major C# minor	 B major G# minor
 F major D minor	 Bb major G minor	 Eb major C minor	 Ab major F minor	 Db major Bb minor	 Gb major Eb minor

Chord Formulas

Major Chords

To find the notes of any major chord begin with your root.
This is the letter name of the chord.

Starting with the note one-half-step above it, count
up 4 half-steps. This is the next note in your chord.

Count up 3 more half steps. This is your third note.

Minor Chords

To find any minor chord, again begin with the root.
Starting with the note one-half step above, count up 3 half steps.
This is the next note in your chord. Count up 4 more half steps.
This is your third note. ie: Cm C-Eb-G

More Chords Formulas

Diminished: Root + 3 half steps (2nd tone) + 3 half steps (3rd tone)
ie: Cdim is C-Eb-Gb

Augmented: Root + 4 half steps (2nd tone) + 4 half steps (3rd tone)
ie: C^{aug} = C-E-G[#]

To find the 7th add 3 half steps to any chord ie: C7 = C-E-G-Bb

To find the Major 7th add four half steps ie: Cmaj7 is C-E-G-B

Add 2 half steps to any chord to find the 6th. ie: C6 is C-E-G-A
(It is the chord with its relative minor)

To play a 9th simply play a major chord and add the note 1 whole step above the
root. ie: C9 is C--E-G-D

A suspended adds the 4 of the scale. ie: Csus = C- E- G add F

Chord Symbols

C = major

C- or Cm = minor

Co = C diminished

C+ = C augmented

C6 = add the 6th

C9 = add the 9

C7 = add the 7b

Csus = add the 4th

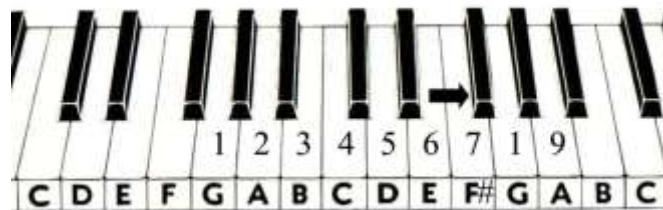
C Δ 7 or CMaj7 =
add the natural 7th

Patterns

As you can see, making music is really about numbers, patterns and putting them all together. Let's take a look at the chords in the C, F,, and G scales to compare the similarities.



C	Dm	Em	F	G	Am
C-E-G	D-F-A	E-B-D	F-A-C	G-B-D	A-C-E
1	2-	3-	4	5	6-

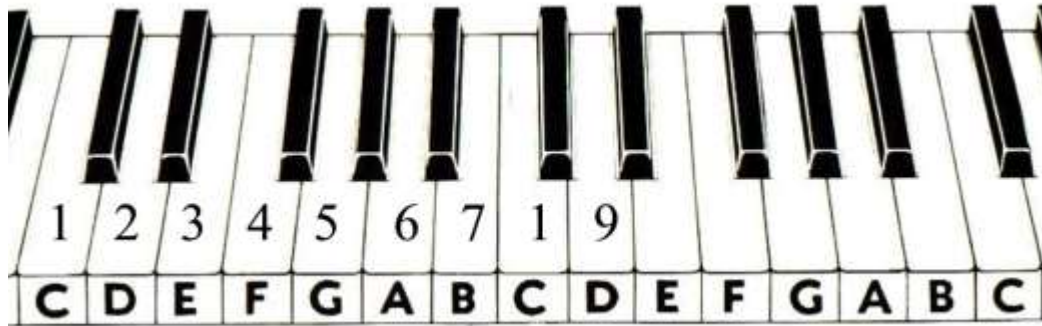


G	Am	Bm	C	D	Em
G-B-D	A-C-E	B-D-F#	C-D-E	D-F#-A	E-G-B
1	2-	3-	4	5	6-



F	Gm	Am	Bb	C	Dm
F-A-C	G-Bb-D	A-C-E	Bb-D-F	C-E-G	D-F-A
1	2-	3-	4	5	6-

Relative Minors



As you learned earlier, the 6th note of the scale is called the relative minor. The relative minor is like a tender, sad or mysterious cousin. The relative minor chord can complement or sometimes even be used instead of the Major chord because both chords share two of the same notes. This happens most often with using the 2m instead of the 4 chord.

For example, Am is the relative minor to C. The notes of a C chord are C-E-G. The notes of an Am chord are A-C-E. Both include the C and the E.

Let's take a look again at the 6 chords we learned in the C Scale.

C Dm Em F G Am

Am is the relative minor to C.

Dm is the relative minor to F

Em is the relative minor to G.

The Relative Minor Scale has the same key signature as its Major Key. That is they both contain the same notes. To play the relative minor scale, simply begin with the 6th note of the major scale.

G scale

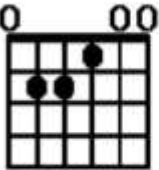
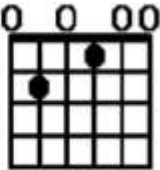
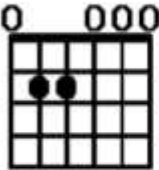
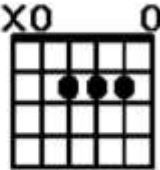
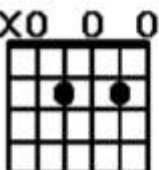
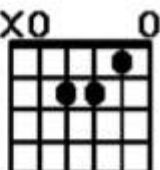
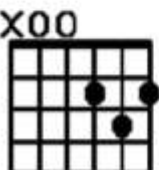
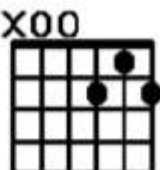
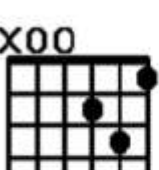
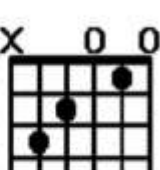
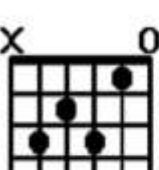
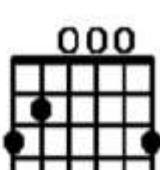
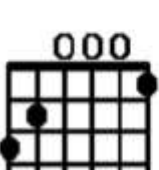
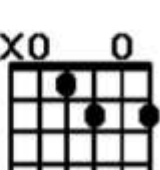
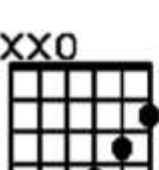
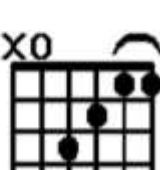
G A B C D E F# G

Em Scale

(E is the 6th note of the G scale)

E F# G A B C D E

Guitar Chord Chart

E  2 3 1	E7  2 1	Em  2 3	A  1 2 3
A7  1 3	Am  2 3 1	D  1 3 2	D7  2 1 3
Dm  2 3 1	C  3 2 1	C7  3 2 4 1	G  2 1 3
G7  3 2 1	B7  1 2 3	Bm  3 2 1	F  3 2 1

I have played thousands of gigs knowing just these chords. The numbers below each chord tell you what finger to use in that position. With guitar your pointer finger is 1, your middle finger 2, your ring finger 3 and your pinky is 4.

Scale Chart

DO	RE	MI	FA	SOL	LA	TI
1	2	3	4	5	6	7
keys C	D	E	F	G	A	B
G	A	B	C	D	E	F#
D	E	F#	G	A	B	C#
A	B	C#	D	E	F#	G#
E	F#	G#	A	B	C#	D#
B	C#	D#	E	F#	G#	A#
F	G	A	Bb	C	D	E
Bb	C	D	Eb	F	G	A
Eb	F	G	Ab	Bb	C	D
Ab	Bb	C	Db	Eb	F	G
Db	Eb	F	Gb	Ab	Bb	C
Gb	Ab	Bb	Cb	Db	Eb	F

Using the Scale Chart to Transpose

1	2-	3-	4	5	6-	7o
C	D	E	F	G	A	B
G	A	B	C	D	E	F#

Until you learn the scales of the keys you like to sing in or if you are playing an instrument for someone who wants to sing in a different key than a piece of music is written in, you can use the Scale with chords chart to make a music chart.

Usually, the last note and chord of the song is the same as the key the song is written in. If it is a minor key song like *Greensleeves*, it will most likely end in the minor key which is the relative minor of the key signature.

Let's make this simple. The chart above illustrates how to transpose a song written in the key of C to the key of G. I like to write the chords over the lyrics because I am a singer. However, as you will see, musicians don't need the lyrics to play a song.

Make 7 columns and 3 rows on a piece of paper. Number the first row from 1 to 7 like the chart above. On the second row, write the letters of the key the song is written in. In the chart above, the song is written in C.

On the third row you will write the letters of the key you want to transpose the song to. Here we are transposing from the key of C to the key of G.

C is the 1 chord in the key of C. 2 is the D chord etc. Simply change all of the Cs to G's, all of the D's to A's, all of the E's to B's etc. **MAKE SURE TO MAKE IT THE SAME KIND OF CHORD!** This means that if it is an Fm in the C Scale make it a Cm in the G scale. A letter standing alone is ALWAYS major.

If you were to see a Bb chord in the Scale of C chart that would transpose to F in the G chart because Bb is half a step lower than B (C chart) and F is half step lower than F# (G Scale).

Chord Progressions

1	2-	3-	4	5	6-	7o
C	Dm	Em	F	G	Am	Bdim
G	Am	Bm	C	D	Em	F#dim
D	Em	F#m	G	A	Bm	C#dim
A	Bm	C#m	D	E	F#m	G#dim
E	F#m	G#m	A	B	C#m	D#dim
B	C#m	D#m	E	F#	G#m	A#dim
F#	G#m	A#m	B	C#	D#m	E#dim
C#	D#m	E#m	F#	G#	A#m	B#dim
F	Gm	Am	Bb	C	Dm	Edim
Bb	Cm	Dm	Eb	F	Gm	Adim
Eb	Fm	Gm	Ab	Bb	Cm	Ddim
Ab	Bbm	Cm	Db	Eb	Fm	Gdim
Db	Ebm	Fm	Gb	Ab	Bbm	Cdim
Gb	Abm	Bbm	Cb	Db	Ebm	Fdim

These are the natural chords found in each scale, meaning every chord is made with the notes in that scale. Sometimes songs have different chords than what you see on this chart. For instance, many country songs use the 2 major instead of minor.

Music Charts

Writing a Chord Chart

Measures hold the value of the rhythm of the song. I like to use the slash to separate the measures but it is not necessary.

Many times the slash marks are not used. If there are two or more chord changes inside of a measure, the measure will be written in parenthesis with slashes over each chord change to indicate the time value each chord is played. (page 38)

4/4 time has a 1-2-3-4 count per measure.

3/4 counts 1-2-3 per measure. This is waltz time.

6/8 counts 1-2-3-4-5-6 per measure but it can be counted out in 4/4 time.
The song *Blueberry Hill* is a 6/8 song.

2/4 counts 2 beats per measure like a polka. *Jose Cuervo* (my song) is in 2/4 time.

A diamond  indicates that you hold the previous chord and not play.

:|| is a repeat sign. It means to go back to this symbol ||: and play the music again.

Chord Symbols

(Refer to page 24 for chord formulas)

Chord Symbol	Chord Name	Example	Intervals
maj	Major	C, Cmaj	1 - 3 - 5
6	Major Sixth	C6	1 - 3 - 5 - 6
maj7, Δ 7	Major Seventh	Cmaj7, C Δ 7	1 - 3 - 5 - 7
min, m, -	Minor	Cm, C-	1 - $\flat$ 3 - 5
m7, -7	Minor Seventh	Cm7, C-7	1 - $\flat$ 3 - 5 - $\flat$ 7
7, dom7	Dominant Seventh	C7, Cdom7	1 - 3 - 5 - $\flat$ 7
dim, $^{\circ}$	Diminished	Cdim, C $^{\circ}$	1 - $\flat$ 3 - $\flat$ 5

(Medium Swing)	All Of Me		Gerald Marks				
A 4/4 F _Δ 7		∕.	A ₇		∕.		
	D ₇		∕.	G ₋₇		∕.	
B 	A ₇		∕.	D ₋₇		∕.	
	G ₇		∕.	G ₋₇		C ₇	
A 	F _Δ 7		∕.	A ₇		∕.	
	D ₇		∕.	G ₋₇		∕.	
C 	B ^b _Δ 7	B ^b _{o7} B ^b ₋₆	F _Δ 7 F _Δ A ₋₇	D ₇			
	G ₋₇	C ₇	F ₆ F [#] _∅	G ₋₇ C ₇			

Let's look at this Jazz chart and compare the chord symbols to the Chord Symbol chart. The slash with the two dots is a repeat sign. This means that when playing the first two measures you will play an Fmaj7 for both of those two measures. Then you will play the A7 for 2 measures, then the D7 etc.

In the C section of the chart, the Bo7 and the Fmaj7 in the smaller font are simply alternative chords you can use.

YOU ARE MY SUNSHINE

1 1 1 17

4 4 1 1

4 4 1 1

1 5 1 1

This is the musical number chart to *You Are My Sunshine*.

Each of the numbers has a value of 4 counts. Pick a key and play the song.
Here is a table of 3 popular keys.

	1	4	5
Key of C	C	F	G
Key of G	G	C	D
Key of D	D	G	A

Note: The D chord has an F# and the A chord has a C#.

Merle Haggard - The Way I Am chords

A

Wish I was A down on some blue G bayou, D
 With a bamboo cane stuck in the sand. A E
 But the road I'm on, don't seem to go there, A G D
 So I just dream, keep on bein' the way I am. A E A

A = verse

1	1	(7b 4)	4
1	1	5	5
1	1	(7b 4)	4
1	5	1	1

A

Wish I enjoyed what makes my living, A G D
 Did what I do with a willin' hand. A E
 Some would run, ah, but that ain't like me. A G D
 So I just dream and keep on bein' the way I am. A E A

B = chorus

B

The way I am, don't fit my shackles. D A
 The way I am, reality. E A
 I can almost see that bobber dancin', A G D
 So I just dream, keep on bein' the way I am. A E A

4	4	1	1
5	5	1	1
1	1	(7b 4)	4
1	5	1	1

I copied this chart from the internet. We can see that the song chart is written in A major because the last chord of this song is A. The G is natural in this song, but in the key of A, the G is sharp. Therefore, when writing the number chart you will write 7b because the natural G is one half step lower than G#.

I sing this song in the key of G. F# is the 7th note of the G Scale. Therefore, if I was to play the song in the key of G, I would play an F natural because it is one-half step below the F#.

GREENSLEEVES

6- 5 4 3

6 5 (4 3) 6-

1 5 4 3

1 5 (4 3) 6-

This is the number chart for the song, *Greensleeves* which is also the Christmas hymn, *What Child Is This*. There are 3 beats to a measure which is the tempo of a waltz.

Although *Greensleeves* is actually written in the Relative Minor key, number charts are always in reference to the major key. For example, in the key of C the relative minor is Am which is the first note of the A minor scale. However, the A would still be referred to as the 6 because it is the sixth note of the C scale. C=1 and A=6.

Try playing the song in different keys. If you want to sing it, select a key that is comfortable for you and write your own chart substituting the numbers for the letter names of the chords you will play. Try playing the root octave (the root is the name of the chord) with different inversions of the chord.

I like to play different chord progressions because it usually spawns a brand new melody. I do it oftenly with the song *Greensleeves*, which happens to be a favorite of mine.

GREENSLEEVES (poss. by Henry VIII of England, 1500's.)

Alas, my love, you do me wrong / To cast me off discourteously

For I have loved you well and long / Delighting in your company

(chorus)

Greensleeves was all my joy / Greensleeves was my delight

Greensleeves was my heart of gold / And who but my lady greensleeves

Your vows you've broken, like my heart / Oh, why did you so enrapture me?

Now I remain in a world apart / But my heart remains in captivity (chorus)

I have been ready at your hand / To grant whatever you would crave

I have both wagered life and land / Your love and good-will for to have (chorus)

If you intend thus to disdain / It does the more enrapture me

And even so, I still remain / A lover in captivity / (chorus)

My men were clothed all in green / And they did ever wait on thee;

All this was gallant to be seen / And yet thou wouldst not love me (chorus)

Thou couldst desire no earthly thing / But still thou hadst it readily

Thy music still to play and sing; / And yet thou wouldst not love me (chorus)

Well, I will pray to God on high / That thou my constancy mayst see

And that yet once before I die / Thou wilt vouchsafe to love me

Ah, Greensleeves, now farewell, adieu / To God I pray to prosper thee

For I am still thy lover true / Come once again and love me (chorus)

IMAGINE

A

1	4	1	4
1	4	1	4
4	4	5	5

B

(4 5)	1	(4 5)	1
(4 5)	1	(4 5)	1

Here is the number chart for John Lennon's song, *Imagine*. This is a great song to practice your chord inversions. We will play it in the key of C. C is the 1 chord.

Begin by playing a C octave in your left hand and the C chord in root position (C, E, and G played with your 1, 3, and 5 finger). Play four beats in a row.

Play the 4 chord in 2nd inversion which is F. Play the F octave with your left hand. Play C with your thumb, F with your middle finger and G with your 5 finger. Play four beats again.

When you play the 5 chord which is G, Put your thumb on the B, your pointer finger on the D and your pinky on the G which is 1st inversion.

Play the notes altogether or rock them back and forth with the note played with your thumb, and the two on top.

IMAGINE

by John Lennon

A - Imagine there's no heaven
It's easy if you try
No hell below us
Above us only sky
Imagine all the people
Living for today... Aha-ah...

A - Imagine there's no countries
It isn't hard to do
Nothing to kill or die for
And no religion, too
Imagine all the people
Living life in peace... You...

B - You may say I'm a dreamer
But I'm not the only one
I hope someday you'll join us
And the world will be as one

A - Imagine no possessions
I wonder if you can
No need for greed or hunger
A brotherhood of man
Imagine all the people
Sharing all the world... You...

B - You may say I'm a dreamer
But I'm not the only one
I hope someday you'll join us
And the world will live as one

Typically songs are structured
with either 1, 2 or 3 segments.
Match them to the *Imagine* chart.

A – Verse

B – Chorus or Refrain

C - Bridge

TO MAKE YOU FEEL MY LOVE

Verse

: 1	5	5m	4
4m	1	(2 5)	1 :

Bridge

4	1	(3 4)	1
4	1	2	(5 57)

This song is written by Bob Dylan and is a favorite of mine. You can convert it to any key using the chord chart on page 25. REMEMBER IF THE NUMBER STANDS ALONE IT INDICATES A MAJOR CHORD.

Each box represents a measure of 4 counts. In Nashville, there are usually no boxes or lines. Notice there is a 5m. In the diatonic scale the 5 is a major chord. There is also a 2 and 3 major whereas in the diatonic scale, the 2 and 3 are minor chords. In other words, there are no absolute harmonic rules when it comes to creating music. Each combination produces a unique sound. Sometimes they can sound sour or distorted but it is always the composer's choice.

I like number charts because I like to accompany records. Adele sings this song in A, Bob Dylan does it in Db (C#) and Garth Brooks does it in the key of D. I like to sing it in the key of A. This means that all of the 1s are A's, The 2s are B's, the 3s are C#'s, the 4s are D's, the 5's are E's the 6s are F#'s and the 7s are G#'s.

TO FIND ANY CHORD QUICKLY REFER TO THE FORMULA ON PAGE **.

:|| This sign means to loop back to this sign ||: and repeat what you just played.

TO MAKE YOU FEEL MY LOVE

Bob Dylan

(Verse)

When the rain is blowing in your face
And the whole world is on your case
I could offer you a warm embrace
To make you feel my love

Verse

A / E

Em / D

Dm / A

(B E) / A

(Verse)

When the evening shadows and the stars appear
And there is no one there to dry your tears
I could hold you for a million years
To make you feel my love

Bridge

D / A

(C# D) / A

D / A

B / E

(Bridge)

I know you haven't made your mind up yet
But I would never do you wrong
I've known it from the moment that we met
No doubt in my mind where you belong

(Verse)

I'd go hungry, I'd go black and blue
I'd go crawling down the avenue
No, there's nothing that I wouldn't do
To make you feel my love

(Bridge)

The storms are raging on the rolling sea
And on the highway of regret
The winds of change are blowing wild and free
You ain't seen nothing like me yet

(Verse)

I could make you happy, make your dreams come true
There's nothing that I wouldn't do
Go to the ends of the Earth for you
To make you feel my love

(Tag) To make you feel my love

(This is my chart of this song)

MOON RIVER

This is a sample of a chord and lyric sheet that you can find online. Let's turn it into a number chart. Each measure will have 3 counts. This song is in the key of G, therefore G will be our 1, A will be 2, B will be 3, C will be 4, D will be 5, E will be the 6 and F# will be the 7.

G Em Cmaj7 G

Moon River wider than a mile

Cmaj7 G F#m B7

I'm crossing you in style some day

Em Dm Em/D Em

Oh dream maker you heart breaker / //

Em F# (Bm E) (Am D)

Wherever you're going I'm going my way

G Em Cmaj7 G

Two drifters off to see the world

Cmaj7 G F#m7 B7

There's such a lot of world to see

Em Dm Em/C# Cmaj7

We're after the same rainbow's end

G C#7 G

Waiting round the bend

Cmaj7 Bm Em Am D G Cm7 G

My Huckleberry Friend Moon River and me

MOON RIVER

G	Em	Cmaj7	G
1	6m	4maj7	1
Cmaj 7	G	F#m	B7
4maj7	1	7#m	37
Em	Dm	Em/D	Em
6m	5m	6m/5	6m
Em	F#	(/Bm //E)	(/Am //D)
6m	7	3m	(2m 5)
G	Em	Cmaj7	G
1	6m	4maj7	1
Cmaj7	G\	F#m7	B7
4maj7	1	7m7	37
Em	Dm	Em/C#	Cmaj7
6m	5m	6m/4#	4maj7
G	C#7	G	Cmaj7
1	4#7	1	4maj7
Bm	Em	Am	D
3m	6m	2m	5
G	Cm7	G	G
1	4maj7	1	1

FLY ME TO THE MOON

A	6-7	2-7	5 ₇	1maj7
	4maj7	7 _{o7}	3 _{7b9}	(6- 6 ₇)
	2-7	5 ₇	(1maj7 4 ₇)	(3-7 6 ₇)
	2-7	5 ₇	1maj7	(7 _{o7} 3 _{7b9})
B	6-7	2-7	5 ₇	1maj7
	4maj7	7 _{o7}	3 _{7b9}	(6-7 6 ₇)
	2-7	5 ₇	3-7	6 ₇
	2-7	5 ₇	1maj7	(7 _{o7} 3 _{7b9})

This is the number chart for *Fly Me To the Moon*. The jazz chart is written in the key of F, and each measure has a 4 count value. The circle with a dash indicates a diminished chord and the triangles with the 7s indicate a maj7. The flat before the 9 means to play the chord with the tone one-half step below the 2 chord. Compare the two with the chart on pages 25 and 26.

(Medium Swing)

Fly Me To The Moon

Bart Howard

A

$\frac{4}{4}$ $\parallel$ D_{-7} | G_{-7} | C_7 | $F_{\Delta 7}$ |

| $B_{\Delta 7}^b$ | $E_{\emptyset 7}$ | A_{7b9} | D_{-7} D_7 |

| G_{-7} | C_7 | $F_{\Delta 7}$ B_7^b | A_{-7} D_7 |

| G_{-7} | C_7 | $F_{\Delta 7}$ | $E_{\emptyset 7}$ A_{7b9} $\parallel$

B

$\parallel$ D_{-7} | G_{-7} | C_7 | $F_{\Delta 7}$ |

| $B_{\Delta 7}^b$ | $E_{\emptyset 7}$ | A_{7b9} | D_{-7} D_7 |

| G_{-7} | C_7 | A_{-7} | D_7 |

| G_{-7} | C_7 | $F_{\Delta 7}$ | $E_{\emptyset 7}$ A_{7b9} $\parallel$

I WALK THE LINE

© 1956 (Renewed 1984) HOUSE OF CASH, INC. (BMI)/Administered by BUG MUSIC

Words and Music by
JOHN R. CASH

Moderately bright

1. I keep a close watch on this heart of mine. I keep my eyes wide
 2. very very eas - y to be true. I find my - self alone
 3.-5. (See additional lyrics)

o - pen all the time. I keep the ends out for the tie that binds.
 when each day is through. Yes, I'll ad - mit that I'm a fool for you.

Be - cause you're mine I Walk The Line. I find it Line.
 Be - cause you're mine I Walk The Line. As sure as

Additional Lyrics

3. As sure as night is dark and day is light,
 I keep you on my mind both day and night.
 And happiness I've known proves that it's right.
 Because you're mine I Walk The Line.

4. You've got a way to keep me on your side.
 You give me cause for love that I can't hide.
 For you I know I'd even try to turn the tide.
 Because you're mine I Walk The Line.

5. I keep a close watch on this heart of mine.
 I keep my eyes wide open all the time.
 I keep the ends out for the tie that binds.
 Because you're mine I Walk The Line.

Here is an example of what a song looks like in a Big Fake Book. As you can see the song is written in the key of F (one flat) and there are 4 beats to each measure. Although many songs begin in the key they are written in, this one doesn't. It is the last chord of the song that tells us what key it is in.

Notice there is a repeat sign $\text{II}:$. The half box over the word "Line" are endings. The 1-4 mean that you play that ending after verses 1 to 4. The one over the 5 you play after the 5th verse. The double line at the end indicates the end of the song.

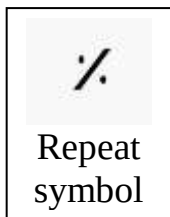
This is the number chart for this song. Look and compare to the music above. I put 3 slashes on the second line to show me that the 17 is played on the 4th beat. You can also write that measure as shown in the box. The slashes = counts.

/	/	/	/
(1		17)	

II: 57	57	1	1
57	57	1	///17
4	4	1	1
57	57	1	1 :II

CHORD CHART WRITING REVIEW

1. To determine how many counts per measure:
 4/4 time is the most popular
 3/4 waltz
 2/4 polka
 6/8 *Blueberry Hill*
2. Not all song begin with the 1 chord. To determine the key you are looking to transpose or write a number chart it is usually the last chord of a song.
3. A letter or number standing alone is always Major.
4. Many times songs do not use chords consistently from its original diatonic scale. In transposing and writing number charts, the chords must be duplicated exactly.
5. When writing a chart, if a chord is made sharp (#) or flat (b) in a piece of music that is not consistent with the natural diatonic scale, alter the chord accordingly. That is one half step up for sharps and one half step down for flats.
6. Whenever you see a letter or number with a slash and then another letter or number, the first letter names the chord and the second letter or number names the bass note. Example: G/C in the C Scale is also 5/1.



Chord Symbol	Chord Name	Example	Intervals
stands alone	Major	C	1 - 3 - 5
m or -	Minor	Cm or C-	1 - 3b - 5
7	Seventh	C7	1 - 3 - 5 - 7b
m7 or -7	Minor Seventh	Cm7 or C-7	1 - 3b - 5 - 7b
Maj 7 or Δ 7	Major Seventh	Cmaj7 or C Δ 7	1 - 3 - 5 - 7
dim or o	Diminished	Cdim or Co	1 - 3b - 5b
Aug or +	Augmented	Caug or C+	1 - 3b - 5#
9	Ninth	C9	1 - 3 - 5 - 9
6	Sixth	C6	1 - 3 - 5 - 6

Interesting Musical Facts

“It is the transparency, the arches and rhythms between the open space that so profoundly stir the human spirit.” ~ The Mozart Effect

Legend says that Pythagoras brought a tangible understanding of music when he walked by a blacksmith and realized two different weighted hammers hit together sometimes produces a harmonic tone. This evolved into creating the harp which evolved into a piano. The length of the strings is in perfect ratio.



The octave relationship is a natural phenomenon that has been referred to as the “natural miracle of music.” The song “Over the Rainbow” starts with an octave. In an octave the lower note of the octave vibrates exactly one-half the amount of hertz as the higher note.

The Pythagorean scale is called the Diatonic Scale and is based on the 3:2 ratio. We know it as Do-Re-Mi-Fa-Sol-La-Ti- Do. The perfect fifth is 5 notes above the tonic (letter name of the scale) and the fourth is 5 notes below.

The Pentatonic scale is basically the 5 black notes on the keyboard. It is sometimes referred to as “The Slave Scale” because black spirituals are written with these five notes. Amazing Grace was written on a slave ship. You can play the melody on the black keys beginning with C#.

Piano music is healthy because it ranges from 27.5 to 4168 hertz. This is the measurement of vibration for each tone the bigger numbers being the higher tones. The high notes stimulate the mind, the middle tones the solar plexus and emotions. The low tones on the piano make us sleepy when played slowly and stimulate us to move when played fast.

Brain waves, heart rhythms, blood pressure, organic rhythms, and emotional tone can subtly change according to the music we listen to. When deeply relaxed the brain releases beta-endorphins.

The ability of music to influence human emotion is well known, and is used extensively by moviemakers. A variety of musical moods may be used to create feelings of calmness, tension, excitement, or romance. Lullabies have long been popular for soothing babies to sleep. Music can also be used to express emotion nonverbally, which can be a very valuable therapeutic tool.

Depending on the type and style of sound, music can either sharpen mental acuity or assist in relaxation. Memory and learning can be enhanced, and this used with good results in children with learning disabilities. This effect may also be partially due to increased concentration that many people have while listening to music. Better productivity is another outcome of an improved ability to concentrate. The term “Mozart effect” was coined after a study showed that college students performed better on math problems when listening to classical music.

Music Quotes

"Music directly imitates the passions or states of the soul... when one listens to music that imitates a certain passion, he becomes imbued with the same passion; and if over a long time he habitually listens to music that rouses ignoble passions, his whole character will be shaped to an ignoble form. ~ Aristotle

"Music gives soul to the universe, wings to the mind, flight to the imagination, a charm to sadness, gaiety and life to everything; It is the essence of order and lends to all that is good, just, and beautiful." ~ Plato

"Musical training is a more potent instrument than any other, because rhythm and harmony find their way into the inward places of the soul." ~ Socrates

*A painter paints pictures on canvas. But musicians paint their pictures on silence.
~Leopold Stokowski*

*There is geometry in the ringing of strings. There is harmony in the spacing of the spheres.
~Pythagoras*

Music works in mysterious ways. You are amazed at the power of it. ~ Keith Richards

*Whoever controls the media controls the mind. The media is the message and the message is me.
~Jim Morrison*

I can explain everything better through music. You hypnotize people and preach into their subconscious what we want to say. ~ Jimmy Hendrix

I realized that music got through to youngsters because the beat matches the natural rhythms of the body. ~ Frank Zappa

Never was there an art form directed so exclusively to children...the music business is peculiar only in that it caters almost exclusively to children.~ Dr. Alan Bloom

Scientists are finding that the human brain is pre-wired for music. The brain seems to be a sponge for music and like a sponge in water, is changed by it. ~ Newsweek Magazine

Where words fail, music speaks.~ Hans Christian Andersen

One good thing about music, when it hits you, you feel no pain. ~Bob Marley

If music be the food of love, play on. ~ William Shakespeare

Music is the mediator between the spiritual and the sensual life. ~Ludwig van Beethoven

Cynthia Jordan

“to heal the world with beautiful music”

Creative, talented, fun and entertaining, speaker/composer/author, Cynthia Jordan, is dedicated to her vision of bringing awareness to the power of music and she is right on track. “Music used with intention can make a dramatic difference in people’s lives.” With the success of her music compositions and now historical novels and new musical, Pearl, Cynthia likes to call herself a story-teller. Her work has proven to be memorable as well as life-changing.



In 1983, every country radio station and honky-tonk was playing, Jose Cuervo you are a friend of mine.” This is the first song this California surfer girl ever wrote! The song Jose Cuervo went on to be the 1983 Country Music song of the year in Billboard Magazine. That same year Jose Cuervo tequila sales increased 14 million cases. “That is the power of music.”

Music is in Cynthia’s family. Her great-grandfather Alberto Solis was a talent Scout for Irving Berlin and worked in the earliest years for RCA Victor talking machines. Her paternal grandmother, Opal Jordan was a honky-tonk piano player. After losing her husband in WWII, Opal raised her four children working in ammunition factories during the day and playing music at night. “My grandmother was definitely one of the last of the red hot mamas!”

As fate would have it, Cynthia moved to Nashville, Tennessee where she acquired a contract to compose and produce 11 CDs for Page Music. She created The Peaceful Journey Series, 4 Christmas CDs and a 3 CD project she named New Beginnings for pregnant women. Within two years Cynthia had over 5 million downloads of her soothing piano music and began receiving letters from all over the world.

“We use your music to relieve our little girl’s migraine headaches.”

“I had my first full night’s sleep in two years!”

“We played your Celtic CD when we buried our son and it brought us great comfort.”

Cynthia Jordan was excited to learn there has been much research on the power of music. In 1998 she began her crusade to demonstrate how music can improve people’s lives. In 2005 Cynthia moved to West Texas where she hosts a 3 hour pre-recorded radio program on 97.1 FM she calls, Ahh Radio with the intention to sedate her listeners.

Currently her new historical novels, Pearl and Diamond AND Ruby have created a new crusade discussing women’s challenges that exploded in the 1920’s. To date Cynthia has written 3 theatrical musicals, Lady in Blue, Cowboy Cabaret and Pearl, which is full of songs that make her audiences want to dance in the aisles.

More from Cynthia



Non- Fiction and Historical Novels



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